1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

High tumor budding density at the invasive front (BD-IF) is an independent predictor of shorter progression-free interval (PFI) in non-muscle-invasive bladder cancer (NMIBC), after adjusting for standard clinicopathologic factors (tumor grade, stage) and tumor-stroma context (TSR). BD-IF reflects an EMT-like local invasion phenotype that facilitates microinvasion and seeding of urothelial tissue, thereby increasing the risk of local progression and recurrence. The BD-IF metric, when quantified using standardized digital pathology (with predefined field selection, budding definitions, and automated/semi-automated counting) across centers, will improve risk stratification beyond existing models (e.g., risk tables based on grade/stage) and will yield meaningful reclassification for intensified surveillance or adjuvant therapy choices. The hypothesis will be tested by: (1) assembling retrospective multi-center NMIBC cohorts with long-term follow-up; (2) applying a pre-specified BD-IF scoring protocol, ideally with automated image analysis to measure bud density per invasive-front length or per high-power field; (3) evaluating BD-IF as an independent variable in multivariable Cox models for PFI, adjusting for grade, stage, TSR, age, and other covariates; (4) assessing incremental prognostic value using time-dependent AUC, C-index, NRI, and IDI; (5) validating findings in an external cohort; and (6) exploring correlations with EMT-associated morphological features as supportive, non-mandatory evidence. The approach aims for a robust, implementable biomarker that can be integrated into routine pathology reporting and clinical decision-making.

Element	Value
Imaging feature	Tumor budding density at the invasive front (BD-IF)
Primary endpoint	PFI

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Pathologic grade	Supporting modifier from knowledge graph
Secondary	Clinical stage	Supporting modifier from knowledge graph

Secondary	Tumor-stroma ratio (TSR) as a contextual stromal modifier	Supporting modifier from knowledge graph
Secondary	Measurement reproducibility/automation (standardized BD-IF quantification)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Pathologic grade: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Clinical stage: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.